Nirmal Gopalakrishnan

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PROFILE

I am a highly productive and organized individual striving to provide the best service possible to my company. With a strong proficiency in various programming languages and technologies, I excel at quickly learning and adapting to new software and systems. My educational background in computer science, paired with hands-on project experience, equips me with the skills needed to tackle complex challenges in tech development

EDUCATION

KATHIR COLLEGE OF ENGINEERING, — 2022 – 2026 | Coimbatore
Bachelor of Engineering Computer Science and Engineering
CGPA: 7.1

Gandhi Matric Higher Secondary School, *HSC* — 2021 – 2022 | Namakkal
Percentage : 65

Gandhi Matric Higher Secondary School, *SSLC* — 2019 – 2020 | Namakkal
Percentage : 89

PROFESSIONAL EXPERIENCE

Clover Technology, *Game Developer Intern* — 08/2025 – 01/2026 | Coimbatore
Trained in **Blender** and **Unreal Engine**, creating 3D assets and game-ready environments. Gained hands-on experience in 3D modeling, texturing, and animation, along with integrating assets into game engines. Strengthened practical skills in building visually engaging and optimized 3D game content.

IPCS Global Pvt Ltd, *Data Analytics Intern* — 06/2025 – 07/2025 | Coimbatore
A one-month data analytics internship provided practical exposure to data cleaning, analysis, visualization, and basic tools like Excel, Python, SQL, and Power BI, helping develop strong analytical

CORE SKILLS

Game Development (Blender, META HUMAN, 3D Basics) | **Technologies** (Web Development, Data Structures, Algorithms) | **Languages** (Python, Java, C) | **Concepts** (OOPS, Secure Design, Cloud Computing)

CERTIFICATES

Introduction to AI and Vector Search (Dec 2023) | **TensorFlow 2.0 for deep learning** (Infosys • Dec 2025) | **Introduction to Neural Networks** (Simpli Learn • Dec 2025) | **Oracle cloud infrastructure** (Oracle) | **Java programming** (Smart Yugam Academy • Dec 2025) | **DBMS** (Infosys)

PROJECTS

Deep Learning Based Stuttering Detection in Speech Signals Using CNN and Bidirectional LSTM 04/2026

- Designed and implemented a hybrid CNN–BiLSTM model to accurately detect stuttering patterns such as repetitions, prolongations, and speech blocks.
- Applied MFCC-based feature extraction to capture critical acoustic characteristics of speech signals.
- Integrated speech-to-text processing using advanced ASR techniques for converting audio input into textual form.
- Developed an NLP-based pipeline to refine disfluent text into grammatically correct and meaningful sentences while preserving context.
- Implemented text-to-speech (TTS) to reconstruct fluent and natural-sounding speech output.
- Built an interactive diagnostic dashboard to visualize speech patterns, confidence scores, and analysis results.
- Designed the system to support real-time and offline processing, enhancing usability in practical scenarios.
- Technologies Used: Python, TensorFlow, Keras, Librosa, OpenAI Whisper, NLP (spaCy, NLTK), Flask

Al-Powered Personalized Recommendation Engine, 04/2025

Mini Project

- Al-Based Recommendation Engine: Developed using collaborative filtering, content-based filtering, and a hybrid model to deliver personalized suggestions similar to Netflix and Amazon.
- Technologies Used: Implemented in Python using TensorFlow, PyTorch, and the Surprise library; deployed via a Flask web API with MySQL/MongoDB for data storage.
- Performance & Results: Achieved a Mean Squared Error (MSE) of 0.89 and received 80% positive user satisfaction based on feedback.
- Future Scope: Plans to integrate advanced models like BERT (for NLP) and Graph Neural Networks (GNN) to improve accuracy and recommendation diversity.

AUTOMATED TELLER MACHINE SYSTEM, *Java Mini Project* 01/2025

- Engineered an ATM system in Java implementing Object-Oriented Programming (OOP) principles.
- Developed core functionalities: balance inquiry, cash withdrawal, PIN change with secure authentication.
- Applied abstraction and encapsulation via ATM (abstract class) and BankATM classes to ensure data security.

Conducted unit testing with ATMSimulation to validate functionality.

- Technologies: Java, OOP, Secure Design, Software Testing

FILE SHARING AND STORAGE SYSTEM, *Mini Project* 08/2024

- Designed and implemented a cloud-based file-sharing and storage system enabling secure public and private access.
- Developed multi-client architecture allowing users to store and retrieve files efficiently.
- Integrated encryption protocols for secure file transfers
- Technologies: Cloud Computing, Distributed Storage, Secure File Transfer

LANGUAGES

English | Tamil